

Markscheme

Mathematics: analysis and approaches
Practice question bank

88. (a)

Consider the statement: "for every positive integer n , the expression $2^n + 1$ is a prime number."

Find the smallest value of n for which this statement is false.

test $n = 1$: $2^1 + 1 = 3$, prime; test $n = 2$: $2^2 + 1 = 5$, prime (M1)

test $n = 3$: $2^3 + 1 = 9 = 3 \times 3$, not prime A1

the smallest counter-example is $n = 3$ A1

[3 marks]